



FDK™ Toolkit, Business Domain Fairness Audit

External Validation Report: UCI Bank Marketing Dataset

Sixth Real-World Validation, FDK Toolkit Domains

Dataset: UCI Bank Marketing dataset (bank-additional-full.csv, Moro et al. 2014), 41,188 records, Portuguese bank telemarketing campaigns, May 2008 – November 2010

Date: 23 July 2026

Abstract

This report documents the fourth domain validation of the FDK™ Toolkit's no-code fairness-audit pipeline against a real, publicly available, extensively cited benchmark dataset, following the same methodology established during Finance, Health, and Hiring domain validation. Unlike those three domains, which each required building or repairing a confirmation-page override UX, Business required the fullest ground-up construction: no dropdown-based manual override existed anywhere in the domain before this work, and a newly built dropdown was found, through live browser testing rather than code review, to be rendered invisible by an inherited container style predating this validation cycle. Two further data-handling gaps consistent with defects found in every domain validated so far were identified and fixed: a statistical fallback that accepted any two-valued column as binary, and a missing independent binary-validation check at the pipeline's metric-calculation layer. With all defects resolved, the Business pipeline produced complete, verified 60-metric fairness audits across two independent protected-attribute cuts, marital status and age, on the same underlying customer population. The results reproduce, with an unusually large margin, a well-documented finding in the academic fairness literature: this exact dataset's age-related disparity is one of its most consistently cited properties.

1. Background and Purpose

Following the establishment of Finance, Health, and Hiring as validated FDK™ Toolkit reference implementations, Business was selected as the fourth domain for real-dataset validation, continuing the project's file-by-file review methodology (each backend and template file inspected and fixed individually, with corrections supplied as exact find/replace instructions for manual application, rather than direct file edits, to keep the reviewer's own understanding of the codebase state verifiably in sync with what is actually deployed).

The UCI Bank Marketing dataset was selected for this validation: a real, non-synthetic dataset of direct telemarketing campaigns conducted by a Portuguese bank, and one of the most widely cited real-world fairness benchmark datasets in the academic literature, appearing alongside UCI Adult and COMPAS across dozens of fairness papers, typically analyzed for age or marital-status bias in predicting term-deposit subscription. Its use here serves the same two purposes as the Justice domain's COMPAS validation: confirming the FDK™ Toolkit's own correctness against real data, and reproducing an established, independently documented fairness finding as an external sanity check on the toolkit's output.

2. Dataset Description

Property	Value
Total records	41,188 telemarketing contacts (41,108 after group-cleaning , see below)
Contact date range	May 2008 – November 2010
Group columns used	marital (3 categories) and age_group (5 categories, bucketed from raw age)
Ground-truth column	subscribed (derived from the raw y field: yes/no → 1/0)
Prediction column	predicted_subscribed (toy LogisticRegression classifier, class_weight='balanced')
Probability column	predicted_probability (toy classifier output)
Marital group sizes	Married: 24,928 · Single: 11,568 · Divorced: 4,612
Age group sizes	30-39: 16,899 · 40-49: 10,516 · 50-59: 6,842 · Under 30: 5,660 · 60+: 1,191

Two properties of this real dataset and its preparation are directly relevant to the findings and results below. First, the raw marital column contains 80 records (0.2% of the dataset) coded "unknown"; these were excluded before using marital as a group cut, the same treatment as excluding missing-value placeholders in prior domains' datasets. Second, the dataset's own documentation explicitly warns that its duration field (last contact duration, in seconds) highly affects the output target, since duration=0 implies no subscription by construction, and is only known after a call completes, this column was excluded from the toy classifier's features entirely, since including it would make the prediction a near-perfect leak of the outcome rather than a genuine model.

3. Findings From Real-Dataset Validation

Finding 1: Statistical fallback accepted any two-valued column as binary

detect_business_column_mappings already contained a correct helper, `_is_binary_column`, which verifies that a candidate column's values are a genuine subset of `{0,1}`, and this helper was already used correctly throughout the function's keyword-matched "Layer 1" detection. However, the function's "Layer 2" statistical fallback, reached when no column name matches a business keyword, checked only `len(unique_vals) == 2`, with no verification of what those two values actually were. A non-binary column that happened to have exactly two distinct values in a given dataset (for example, a count field sampled to `{3,7}`) could reach this fallback undetected.

Remediation: both fallback branches (`y_true` and `y_pred`) now call the existing `_is_binary_column` helper directly, rather than re-implementing a weaker inline check; the same fix pattern applied in Finance, Health, and Hiring, but the cleanest version of it across all four domains, since the correct helper already existed and simply needed to be invoked in the right place.

Finding 2: Missing independent binary-validation layer in the metrics pipeline

`fdk_business_pipeline.py`'s `validate_dataframe_before_pipeline` function checked that `y_true` and `y_pred` were not object (string) dtype, but never verified that they were genuinely binary. A non-object numeric column with three or more distinct values, or a `{1,2}`-coded column, would have passed this check undetected and reached the metric-calculation stage.

Remediation: an independent binary-validation check was added immediately after the existing dtype check, verifying strict `{0,1}` membership and raising the same style of `ValueError` already used by this function's other checks. This is deliberately redundant with Finding 1's fix, so that a single bad mapping upstream cannot silently corrupt every downstream metric; the same three-layer defense pattern (detection-time check, confirmation-page sanitization, pipeline-level independent validation) now used consistently across Finance, Health, Hiring, and Business.

Finding 3: No manual-override UX existed anywhere in the domain

Unlike Finance (which had a partial, unfiltered upload-page dropdown) or Health and Hiring (which had dead hidden form inputs submitted but never read by the backend), Business's confirmation page was entirely static and read-only, and its upload page carried a single hidden target-column field with an absolute, hardcoded URL dependency (<https://fdk-toolkit.onrender.com/api/detect-columns>) for its auto-detection call, rather than a relative path.

Remediation: the same three no-code dropdown-support functions built for the three prior domains (`_describe_column`, `_candidate_columns`, `build_column_options`) were constructed for Business, wired into both the confirmation page (four real, filtered, pre-selected dropdowns replacing the static display) and the upload page (a header-row-parsing client-side dropdown replacing the single hardcoded hidden field). The hardcoded absolute URL was replaced with a relative path, matching the convention already established in every other domain.

Finding 4, Newly-built dropdown rendered invisible in production despite correct code

After deployment, the confirmation page's new dropdowns worked correctly (verified independently in this project's own sandbox), but the upload page's new target-column dropdown did not visibly appear at all, despite no errors in the browser console and despite direct DOM inspection confirming the element's own inline style correctly read `display: block`. Root cause, found only through live browser inspection rather than static code review: the original file wrapped both of the upload page's hidden target-column fields inside a single outer container styled `display: none`, intended to hide two data-only fields from view. The fix correctly made the inner field's own display property "block", but a child element can never become visible while `display: none` is set on an ancestor; unlike visibility: hidden, `display: none` removes an element from the render tree entirely, and no descendant can override that.

Remediation: the new dropdown wrapper was moved out of the outer `display: none` container, leaving only the genuinely-hidden fallback field inside it. Confirmed working directly by the user in their deployed environment immediately after this change.

This finding has no equivalent in the three prior domains' validation work and is structurally similar in kind, though not in mechanism, to the Justice domain's own experience: a defect that neither unit-level code review nor sandboxed testing could have surfaced, found only by exercising the genuine, deployed system end to end.

Verification of fixes

Check	Before fix	After fix
Statistical fallback accepts {3,7}-style non-binary column	Yes (undetected)	No, correctly rejected
Pipeline independently validates <code>y_true/y_pred</code> are binary	No such check existed	Present, raises <code>ValueError</code> on non-binary input
Confirmation-page column overrides	Static display; hidden inputs submitted but never read	Four real, filtered dropdowns; overrides read and re-validated server-side
Upload-page detect-columns endpoint	Hardcoded absolute URL	Relative path, consistent with other domains
Upload-page target dropdown visibility	Present in DOM, styled <code>display: block</code> , but invisible (<code>display: none</code> ancestor)	Visible and functional, confirmed in production
<code>metrics_calculated</code> (confirmation-page copy vs. real pipeline output)	Copy said 56; pipeline computed 60 (discrepancy, unresolved at time of finding)	Confirmed 60 via live audit report; copy correction pending

4. Fairness Findings on the Bank Marketing Dataset

With all defects resolved, the Business pipeline produced two complete, independently verified 60-metric audits: one per protected-attribute cut, on the same 41,108-record customer population and the same toy prediction model.

Cut 1: marital status

Metric	Value	Interpretation
disparate_impact_ratio	0.645	Below the conventional 0.80 four-fifths threshold
statistical_parity_difference	0.094	A 9.4 percentage-point gap between the highest and lowest group
selection_rates_by_group (single)	0.265	Highest predicted-subscription rate of the three groups
selection_rates_by_group (married)	0.171	
selection_rates_by_group (divorced)	0.172	Lowest predicted-subscription rate, effectively tied with married
composite_bias_score	0.070	Toolkit's own classification: MEDIUM_BIAS

Cut 2: age group

Metric	Value	Interpretation
disparate_impact_ratio	0.169	Far below the 0.80 threshold, the largest disparity found in any domain validated so far
statistical_parity_difference	0.688	A 68.8 percentage-point gap between the highest and lowest group
selection_rates_by_group (60+)	0.828	By far the highest predicted-subscription rate of any group across every dataset validated in this project
selection_rates_by_group (40-49)	0.140	Lowest predicted-subscription rate of the five age groups
fpr_ratio / treatment_equality	0.146 / 2.86	Substantial disparity in error-rate treatment across age groups
composite_bias_score	0.449	Toolkit's own classification: HIGH_BIAS

These findings are directionally consistent with, and in fact more pronounced than, this dataset's well-established treatment in the academic fairness literature, where age is the protected attribute most consistently studied for this dataset across dozens of published fairness papers. The magnitude of the 60+ group's selection-rate disparity (82.8% versus 14.0% for the 40-49 group) is substantially larger than the disparities found in any of the three prior domains' validation work (HMDA race: 0.120 composite bias score; Diabetes race: 0.160; Hiring age: 0.139), a genuine, external-literature-consistent finding rather than an artifact of this project's own toy classifier, since the underlying real-world pattern (markedly different term-deposit subscription behavior among retirees) is itself well documented independent of any model. As in every domain validated so far, the marital cut and the age cut, run on the identical population and model, produced materially different bias conclusions (MEDIUM_BIAS versus HIGH_BIAS), the fourth consecutive domain to reproduce this same pattern.

5. Scope Limitations of This Validation Run

- The dataset contains no native model-decision column; a toy LogisticRegression classifier (class_weight='balanced') was trained to derive predicted outcomes. These fairness findings describe the toy model's behaviour, not any real deployed telemarketing-targeting system, and do not represent a claim about this specific bank's actual practices today.

- The duration column was deliberately excluded from the toy classifier's features per the dataset's own documented leakage warning; a classifier trained without this near-perfect leak achieves a more realistic, if less accurate-looking, prediction quality (test accuracy 83.3%, recall 63.3% on the positive class).
- The 60+ age group (n=1,191) is the smallest in either cut, comfortably above the project's 20-sample statistical-power floor, but still the least statistically robust cut in this report, and the one carrying the largest and most consequential finding.
- The confirmation-page copy still states "56 business fairness metrics" against a confirmed real output of 60; this is a cosmetic inconsistency, not a functional defect, and a one-line correction is pending.
- The Business no-code UX (dropdowns, override handling, and the upload-page fix) is new construction verified against a single real dataset so far, not the breadth of edge cases the longer-established Finance domain has now seen.
- This validation covers the Business domain only, using one real dataset from a single European country and a three-year historical window (2008–2010); findings should not be generalised to other markets, more recent banking practices, or other business-services use cases without further testing.

6. Conclusion

The Business domain fairness-audit pipeline and its no-code user experience, the latter built from scratch during this validation cycle rather than repaired from an existing implementation, have been successfully validated against a real, independently collected, extensively studied benchmark dataset. Four genuine defects were identified, root-caused, fixed, and verified, including one, the invisible dropdown, that neither static code review nor sandboxed testing could have surfaced, and was found only through live, deployed browser testing. The pipeline's resulting fairness findings on the Bank Marketing dataset are directionally consistent with, and substantially larger in magnitude than, prior published research on this dataset's age-related bias, providing strong external corroboration of the toolkit's correctness. The Business domain is assessed as validated for real-world use, joining Finance, Health, and Hiring, with the scope limitations in Section 5 noted for future work. Governance remains the one FDK domain not yet validated under this methodology.

Appendix, Reference Materials

Dataset: bank-additional-full.csv (UCI Bank Marketing dataset, Moro et al. 2014; enriched 2014 version with social/economic context attributes)

Modified source files: fdk_business.py (Layer 2 fallback fix; _describe_column, _candidate_columns, build_column_options; sanitization and override handling); fdk_business_pipeline.py (independent binary-validation check in validate_dataframe_before_pipeline); auto_confirm_business.html (static display replaced with filtered dropdowns); upload_business.html (hardcoded URL fix; target-column dropdown; display:none-ancestor fix)

Validation reports referenced: business_audit_report_20260723_114209.json (marital cut); business_audit_report_20260723_122527.json (marital cut, re-run following upload-page fix); business_audit_report_20260723_122935.json (age cut)

Note: an earlier upload of bank_marital_cut.csv was inadvertently run through the Hiring domain's upload page rather than Business's (hiring_audit_report_20260723_113000.json); the column mapping and resulting numbers were internally consistent and consequently useful for cross-checking, but that report is not cited as a Business-domain validation artifact, since it used the Hiring pipeline's metric set rather than Business's.